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Introduction

The compact, high-performance 2804 three-phase brushless DC motor engineered for advanced motion control. It integrates a high-resolution AS5600 magnetic encoder, providing 12-bit (0.087°) accuracy directly on the motor. Operating at a rated 12V, this motor delivers 2600 RPM and 300g·cm of torque, all within an ultra-compact Φ34.5mm x 19.5mm aluminum frame. This integrated system is specifically designed for Field-Oriented Control (FOC), enabling exceptionally smooth, quiet, and efficient operation. Its core strengths

Integrated High-Precision Magnetic Encoder

At the core of this motor's precision is the built-in AS5600 magnetic encoder. This component provides 12-bit high-resolution angular position feedback, achieving an accuracy of up to 0.087°. With versatile I2C, PWM, and analog output interfaces, it delivers the critical data required for implementing sophisticated closed-loop control of position, speed, and torque. This integration eliminates the need for separate encoder mounting and calibration, simplifying the design process.

Optimized for Field-Oriented Control (FOC)

The synergy between the brushless motor and the high-resolution encoder makes this unit perfectly suited for FOC. Field-Oriented Control, an advanced motor control algorithm, utilizes this precise feedback to regulate the magnetic field, resulting in remarkably smooth torque delivery, minimal operational noise, and superior energy efficiency compared to traditional control methods. When paired with a compatible driver, such as the SimpleFOC Mini, it provides a seamless platform for developing high-performance motion systems.

Compact and Robust Construction

Engineered for applications where space is at a premium, the motor features an outer diameter of only 34.5mm and a height of 19.5mm. The housing is precision-machined from CNC aluminum with a black anodized finish, ensuring both structural integrity and effective passive heat dissipation. The design incorporates an outer rotor with NdFeB strong magnets and a hollow shaft, offering flexibility for routing wires or integrating other components through the motor's axis.

Broad Compatibility for Rapid Development

To accelerate project development, this motor is designed for wide compatibility with the open-source ecosystem. It seamlessly integrates with the SimpleFOC library, which provides powerful tools for implementing FOC algorithms. This software support extends to major development boards, including Arduino, ESP32, and Raspberry Pi, lowering the barrier for developers and makers to incorporate advanced motor control into their projects.

Applications

- Pro-Grade Camera Gimbals: Achieve buttery-smooth, jolt-free video stabilization.
- Advanced Robotic Joints: Build quiet, high-torque, and precise joints for robotic arms or legs.
- Force-Feedback Devices: Create responsive haptic feedback systems.
- FOC Learning Platform: The perfect motor to master advanced vector control with SimpleFOC.
- Balancing Robots: Get the high-speed feedback and smooth torque needed for stable self-balancing platforms.

Specification

Motor Electrical Parameters

- Applicable Voltage: DC 7.4-16V
- Rated Voltage: 12V
- Rated Current: 0.4A
- Maximum Current: 1A
- Rated Power: 12W
- Rated Speed: 2600 RPM @ 12V

- Torque: Approximately 300g.cm/0.03Nm
- Number of Slots and Poles: 12N/14P
- Number of Pole Pairs: 7
- Winding Resistance: 5.1Ω
- Phase Resistance: 2.3Ω
- KV Rating: 220KV
- Winding Inductance: 2.8mH
- Phase Inductance: 0.86mH
- Magnetic Flux: 0.0035Wb
- Winding Method: Delta Connection

Motor Mechanical Dimensions

- Stator Diameter: 28mm
- Stator Height: 4mm
- Shaft Diameter: 8mm
- Hollow Shaft Aperture: 5.4-6.5mm
- Single Motor Dimensions: 34.5x19.5mm
- Operating Temperature: 0-60°C

Encoder Parameters

- Operating Voltage: 3.3V
- Chip Model: AS5600
- Resolution: 12-bit
- Accuracy: 0.087°
- Output Method: I2C, PWM, Analog Output (Default: I2C; PGO pin grounded for PWM/analog output)
- Interface: GH1.25-4Pin

Documents

- [Product wiki]
- [3D Model]

Shipping List

- 2804 3-phase brushless DC motor (with encoder) x1
- MX1.25-4P to DuPont cable x1
- 3-pin motor cable x1

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